TASK VAULT

Employee Task & Workforce Management Platform

PROJECT IDEA • PROBLEM STATEMENT • TECHNOLOGY STACK • FEATURES • BUILD PLAN

Purpose

A practical guide to help you plan, design, build, test, and launch a web-based system that helps businesses assign tasks, track progress, and verify completed work.

Product name: Task Vault

At a glance

Target users	Businesses of different sizes, managers, supervisors, and employees.
Platforms	Responsive website first; installable PWA and offline features later.
Core promise	Every task has an owner, a deadline, a status, and a clear record of its outcome.

1. The idea

Task Vault is a business task and workforce management platform. It allows an employer or manager to create tasks, assign them to employees or teams, set deadlines, follow progress, collect evidence of completed work, and approve the result. Employees can use the same system on a phone or computer.

The product supports both office-based tasks (for example, preparing a report) and physical jobs (for example, inspecting a site or cleaning an office). Physical work can include optional photos, checklists, location details, and completion notes.

> **The key difference from a basic to-do list is the complete work cycle:** assign → perform → submit proof → review → approve or request corrections.

Example interface: employer dashboard

A manager dashboard showing open tasks, overdue items, awaiting-review count, and a recent tasks table (task name, assignee, status, due date). Concept illustration only — the real dashboard should show live data from the database.

Example interface: employee phone view

Mobile “My tasks” screen listing assigned work with status badges (In progress, Due today, Assigned) and an Update task action.

Employee experience

1. Sign in securely.
2. Open My Tasks.
3. Read instructions and deadline.
4. Start the work and update progress.
5. Add photos, documents, checklist results, or notes if required.
6. Submit for review and respond to manager feedback.

2. Problem statement

Problem: Many businesses coordinate work through a mix of verbal instructions, WhatsApp messages, spreadsheets, emails, paper checklists, and phone calls. This can make it difficult to know who is responsible for a task, when it is due, whether it is progressing, and whether the submitted work meets the required standard.

Common pain points

Unclear ownership	Employees may not know exactly who is responsible or what outcome is expected.
Missed deadlines	Managers often discover delays only after following up manually.
Limited proof	A task may be marked complete without photos, documents, checklists, or a review where those are needed.
Scattered communication	Instructions and changes can be lost across different channels.
Manual reporting	Managers spend time chasing updates and preparing status reports.
Poor visibility	It is hard to see overdue work, approval queues, workload imbalances, and recurring bottlenecks.

Proposed solution: Provide one secure platform where work is assigned, tracked, evidenced, reviewed, and reported in a consistent way. The system should reduce manual follow-ups, improve accountability, and help managers identify problems earlier.

Who benefits? Business owners: visibility over operations. Managers: easier assignment and review. Employees: clear instructions and deadlines. Clients (later feature): optional confirmation of selected work.

Initial success measures

Track whether tasks are completed on time, how long approvals take, how often work is returned for correction, how much manual follow-up managers report, and whether businesses are willing to pay to continue using the product.

3. User roles and permissions

Role	Main permissions
Business owner	Owns the company workspace, manages billing and settings, appoints administrators, and can view company-wide information.
Administrator	Manages employee accounts, teams, categories, and company configuration, according to permissions.
Manager / supervisor	Creates and assigns tasks, monitors progress, reviews evidence, requests corrections, and approves work.
Employee	Views assigned tasks, updates allowed statuses, comments, reports blockers, and submits evidence.
Client reviewer (later)	Can view or confirm only specific jobs shared with them; cannot browse internal company data.

> **Security rule:** every company must have a private workspace. Users can access only the companies and records they are authorized to access. Never rely on hiding a button in the browser as the only security control.

4. Main features and MVP scope

Feature	Description
Company workspace	Register a company, set company profile, invite employees, and manage roles.
Task management	Create/edit tasks, set assignee(s), priority, due date, instructions, checklist, attachments, and optional job address.
Status tracking	Use a clear task lifecycle: Assigned, In progress, Submitted for review, Approved, Needs correction, Blocked, or Cancelled.
Proof of work	Allow optional photos, documents, checklist answers, notes, and upload timestamps.
Approval workflow	Manager approves or returns work with a reason; employee can correct and resubmit.
Dashboard and filters	Show open, overdue, in-progress, awaiting-review, and approved tasks; filter by employee, team, date, priority, and status.
Notifications	In-app alerts for assignments, approaching deadlines, review decisions, and corrections.
Activity history	Record important changes with actor and timestamp; avoid silently erasing task history.

5. Task workflow

Use one predictable process for both office tasks and physical jobs. Some tasks may skip evidence or approval if the company does not require it, but the system should make that choice explicit.

Flow: 1 Assigned → 2 In progress → 3 Submitted → 4 Reviewed

Suggested status definitions

Status	Meaning
Assigned	Task is allocated but work has not started.
In progress	Employee has started work and can add updates.
Blocked	Employee cannot proceed and records the reason; manager is notified where appropriate.
Submitted for review	Employee says the work is ready and attaches required proof.
Approved	Authorized reviewer accepts the submitted work.
Needs correction	Reviewer gives feedback; the employee updates and resubmits.
Cancelled	Task is stopped with a recorded reason; history is retained.

Example use cases

Office: A manager assigns a monthly report to an employee, sets a due date, and requests a document upload. The employee submits the report; the manager reviews it and approves or returns it.

Physical job: A supervisor assigns a site inspection with a checklist. The employee completes the inspection, uploads permitted photos, records notes, and submits the task for review.

6. Technologies needed

Recommended starter stack: use the tools you already know and keep the first version manageable. You can change technologies later if usage, cost, or product requirements justify it.

Layer	Recommended technology	Purpose
Frontend	HTML, CSS, JavaScript	Responsive screens, task lists, forms, dashboards, mobile layouts.
Authentication	Firebase Authentication	Sign-up, login, password reset, identity.
Database	Cloud Firestore	Companies, memberships, tasks, comments, statuses, notifications.
File storage	Cloud Storage for Firebase	Task photos, documents, and other evidence; secure file rules required.
Backend	Cloud Functions for Firebase or a Node.js API	Trusted actions, validation, notifications, scheduled reminders, billing hooks.
Hosting	Firebase Hosting or another static host	Publish the website over HTTPS.
Version control	Git and GitHub	Track changes, back up code, collaborate, and manage releases.
Design	Figma or paper wireframes	Plan the screens and user flow before coding.
Testing	Browser dev tools, Firebase Emulator Suite, manual mobile testing	Check usability, access rules, forms, file uploads, and error cases.

Important: Firebase pricing and product limits can change. Review current plans before launch. Avoid putting secret keys, service-account credentials, or privileged admin logic in frontend JavaScript. Configure backups, access controls, file size/type limits, and error logging.

7. Data model and security

Suggested Firestore collections

Collection	Example data
companies	Business name, owner ID, settings, created date, plan.
users	Display name, email, account status, profile fields.
memberships	Company ID, user ID, role, team ID, permissions.
teams	Company ID, team name, manager, members.
tasks	Company ID, title, description, assignee IDs, creator ID, due date, priority, status, evidence requirements.
task_updates	Task ID, author ID, comment, status change, timestamp.
task_evidence	Task ID, uploader ID, storage path, file type, timestamp, caption.
task_reviews	Task ID, reviewer ID, decision, feedback, review timestamp.
notifications	Recipient ID, type, message, read state, related task ID.
activity_logs	Company ID, actor ID, action, target record, timestamp.

Security checklist

- Every company-owned record is tied to a company ID and protected by authorization checks.
- A user cannot grant themselves an owner or administrator role.
- Only authorized users can assign, review, approve, or cancel tasks.
- Storage rules prevent unauthorized access to photos and documents.
- Validate inputs and file type/size; never trust client-provided IDs without checking permissions.
- Keep an audit history for meaningful changes and approval decisions.
- Use HTTPS, protect credentials, minimize personal data, and plan for backups and account recovery.
- Follow applicable privacy law, including South Africa’s POPIA where relevant; explain what evidence is collected and why.

8. Build plan and deliverables

A sensible solo-developer plan is to build one complete workflow before adding advanced features. The schedule below is an estimate and may change with available hours and learning needs.

Phase	Work to do	Definition of done
1. Plan	Confirm requirements, roles, statuses, and first target customer. Draw wireframes.	Feature list and screen designs approved.
2. Foundation	Set up repository, Firebase project, environment config, authentication, company workspace, roles.	Users sign in and only see authorized workspace.
3. Tasks	Build task create/edit/list/detail pages, assignments, filters, deadlines, status changes.	Manager can assign work; employee can update it.
4. Evidence	Add uploads, completion notes, review queue, approve/return workflow.	Full task submission and review cycle works.
5. Dashboard	Add summary cards, overdue filters, in-app notifications, activity history.	Managers can see what needs attention.
6. Test	Test desktop/mobile layouts, permissions, file access, bad inputs, errors, and edge cases.	Critical security and workflow tests pass.
7. Pilot	Onboard a few real users, gather feedback, fix confusing steps, review costs.	Pilot users can complete tasks with minimal help.

Suggested MVP screens

- Landing page and company registration
- Login and password reset
- Employer dashboard
- Employee and team management
- Task list with filters
- Create/edit task form
- Task details and activity timeline
- Employee My Tasks page
- Submit evidence form
- Manager review/approval queue
- Basic reports and settings

Estimated timeline: about 6–10 weeks for a focused MVP for a solo developer working part-time. This is a rough planning estimate; scope, experience, testing, and integrations can change it.

9. Business model and launch strategy

Possible pricing to test (not validated market prices)

Plan	Illustrative monthly price	Possible scope
Free trial	R0 for a limited trial	Small team tests core assignment and review workflow.
Starter	R199	Small business with a limited number of employees and storage.
Growth	R499	More employees, reports, recurring tasks later.
Pro	R999+	Larger teams, advanced controls and support.

Validate pricing with interviews and pilot customers. Estimate database, storage, email/WhatsApp messaging, hosting, support, and payment costs before committing to a price.

Recommended launch approach

Although the long-term product can support all businesses, begin by testing with one or two customer types. For example, a cleaning company may need checklists and before/after photos, while a small office team may need document review and deadlines. Build the common core first, then add industry-specific workflows based on evidence.

Questions to ask potential customers

- How do you currently assign work and tell employees about changes?
- How do you know a job is finished and meets the required standard?
- What tasks are most often late, forgotten, or repeated?
- How much time do managers spend chasing updates each week?
- Which evidence is useful: photos, documents, checklists, signatures, or location?
- Would you test this product with your team? What would make it worth paying for?

10. Features for later versions

Version	Features
Version 2	Recurring tasks, templates, team scheduling, exportable reports, email reminders.
Version 3	Optional GPS check-in, QR code at job sites, client sign-off, offline drafts and later sync.
Version 4	AI-generated summaries, suggested assignments, workload insights, risk alerts.
Advanced	Accounting/payroll integrations, inventory, invoicing, multiple branches, enterprise identity tools.

> Do not build all advanced features before confirming that customers use and value the core workflow.

11. Project start checklist

Use this checklist when you are ready to begin development.

- ■ Write the MVP feature list and explicitly exclude advanced features.
- ■ Choose the first business type to interview or pilot with.
- ■ Sketch the employer dashboard, task detail screen, employee task list, and review queue.
- ■ Create a GitHub repository and commit the initial project.
- ■ Create a Firebase project and configure development and production environments.
- ■ Define company, membership, role, task, evidence, review, and activity data structures.
- ■ Implement authentication and company-level authorization before storing real business data.
- ■ Build task creation and assignment before adding uploads or notifications.
- ■ Add evidence uploads and approval workflow.
- ■ Test using at least two separate company accounts and verify they cannot access each other’s data.
- ■ Test on a phone and a computer, including slow network and failed upload cases.
- ■ Run a small pilot, record feedback, estimate operating costs, and refine pricing.

The first milestone

> A manager can create a task, assign it to an employee, and set a deadline. The employee can see the task, update its status, and submit proof. The manager can review the proof and approve the work or request a correction. All of this happens securely inside the correct company workspace.

If this end-to-end workflow is reliable, you have the foundation of a useful product. Add features only when they solve a clear customer problem.

Product name: Task Vault • Product concept and planning guide • Prepared October 2026